

Kivi: Phonetic Word Memory

"The Words Kivi Keeps" — Sarvam Backend-Focused Full Stack take-home

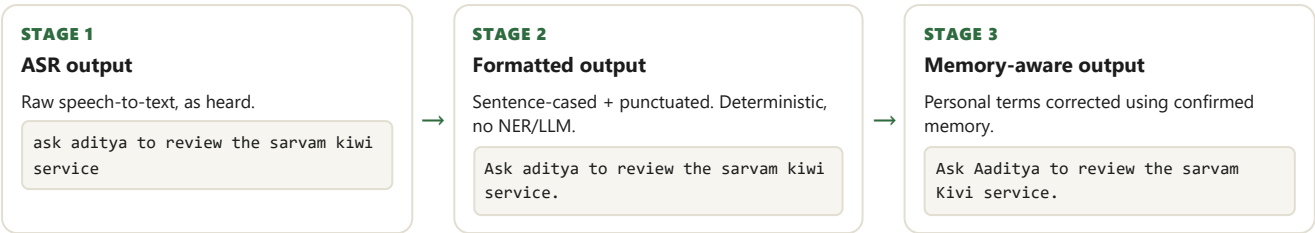
26/26 eval

62/62 tests

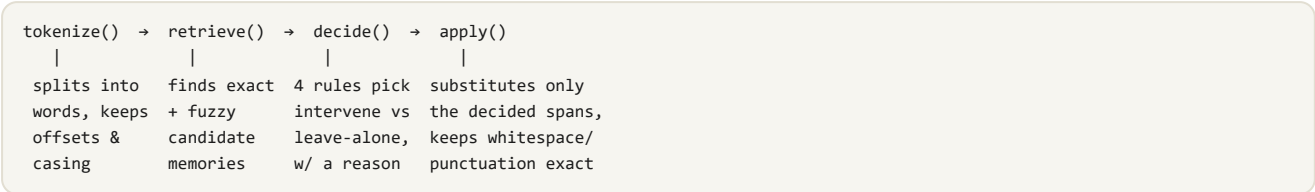
P/R/F1 = 1.0

A word-level memory system for Kivi's dictation pipeline. It learns personal terms (names, products, places) from the corrections a user makes, and applies that memory to fix the same terms later — but only once a correction has been genuinely confirmed twice, never off a single guess.

How it works: three transcript stages



How a transcript gets corrected (Stage 2 → 3, at request time)



The 4 decision rules, in order: **(1)** already correct → leave alone. **(2)** memory not yet `active` (needs 2 confirmations) → leave alone. **(3)** a small hand-curated list of ambiguous common words (e.g. "apple") needs a nearby supporting cue ("stock", "Inc") before applying. **(4)** conflicting active memories for the same span → highest confidence wins. Every decision returns a human-readable `reason` + `explanation`, whether it acted or not.

How a memory gets learned (three independent ways in, one shared core)

PATH	INPUT	MECHANISM
Teach one pair	1 (asr, corrected)	Diff formatted vs. corrected
Bulk pairs	many (asr, corrected)	Same diff, looped
Bulk conversations	correct text only, no ASR	NER (free) + optional LLM refine

Confidence: `min(1.0, evidence_count / 3)`. A memory only applies once `evidence_count ≥ 2` AND `confidence ≥ 0.6` — both cross at exactly the same point, so they can't disagree. One correction is never enough on its own; it could be a typo.

NER + LLM hybrid, verified not assumed

For "corrected text only" input there's no wrong/right pair to diff, so entities must be *extracted*. spaCy NER runs first — free, offline, no API key. An LLM pass refines it *only if* credentials happen to be configured; a missing key is treated as normal, not an error. Real, measured result on one sentence:

ENTITY	FOUND BY	WHY
Kivi, Bengaluru	NER and LLM	Ordinary place/org patterns spaCy recognizes reliably
Aaditya, Sarvam	LLM only	Invented/unusual spellings — exactly spaCy's verified recall gap

This is why the LLM stays a *refinement*, not a requirement: with only the NER model installed (one-time, free download), this mode already works with zero LLM credentials.

What it can tell apart (26 eval cases, beyond the one worked example)

CAPABILITY	CAPABILITY
Needs 2 confirmations before acting	Leaves already-correct text alone
Blocks ambiguous words without supporting context ("apple")	...but allows it once context appears ("Apple" + "stock")
Doesn't over-gate ordinary words not on the ambiguity list ("kiwi")	Handles ALL-CAPS / mixed-case ASR
Handles possessives ("aditya's")	Handles multi-word entities ("yolk sity" → "York City")
Fuzzy-matches small ASR typos within a budget	...but never matches unrelated/very short words

Resolves conflicting evidence toward the most recent	Applies multiple memories correctly in one sentence
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Architecture

```
Browser (frontend/, vanilla JS) -> FastAPI (backend/app.py, routes.py)
+-- POST /observe(/bulk)          -> diff.py          -> memory_service.py -> SQLite
+-- POST /learn-from-conversations -> ner_extraction.py + entity_extraction.py (LLM, optional)
+-- POST /run                     -> formatter -> tokenizer -> retrieval -> decision -> apply
+-- GET/DELETE /memories, POST /reset
```

Two tables: `memory` (one row per learned pair, `UNIQUE(observed_form, canonical_form)`) and `evidence` (append-only log, `source_type` = `observed_pair` or `conversation_extraction` — every memory's full history is inspectable, not just its current state).

Limitations (known and deliberate)

- **Multi-word spans match exactly** — no fuzzy tolerance for spans > 1 token.
- **A taught multi-word entity can fragment** into separate single/multi-word memories, since diffing has no real entity extraction. Mitigated (optional): an LLM grouping pass can merge them when credentials are set.
- **Ambiguous-word stoplist is small and hand-curated**, not a full dictionary — deliberately, to avoid over-gating ordinary product names.
- **No automatic decay** for superseded memories; conflicting evidence competes at retrieval time instead. Manual delete is the "forget" control.
- **Entity-type inference is a coarse heuristic** (nearby cue words), not real NER, in the core diff path.
- **Single-user, SQLite only** — out of scope per the brief's own "smallest system" framing.

Design decisions, in brief

DECISION	WHY
Formatter stays deliberately minimal (no NER)	Assignment scope is phonetic memory, not general formatting — the gap is intentional, memory fills it
Deterministic substitution, not LLM prompt-injection	Part Two needs byte-exact, reproducible evaluation
No Alembic	One schema version for this scope; idempotent <code>CREATE TABLE</code> is enough
No mandatory LLM key anywhere in the primary path	Brief requires zero-credential review; NER hybrid extends this to bulk-conversation import too

Full detail, rationale, and verified findings for every claim above: `README.md` and `RUN.md` in the repository. Built with Claude Code; every number cited here was actually executed and checked in-session, not asserted from memory.